

Aditya Manish Patil

B.E. – Artificial Intelligence & Data Science

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Key Expertise: Machine Learning, PyTorch, ReactJs, Python, Full Stack Web Development, Large Language Models (LLMs)

Professional Summary

Software Engineering student specializing in Artificial Intelligence and Data Science at Dr. D. Y. Patil Institute of Engineering, Management, and Research. Deepening expertise in Machine Learning, LLMs, and Reinforcement Learning, with a focus on mastering core concepts and applying them through hands-on, real-world projects. Experienced in full-stack web development as a medium for implementing ideas, while actively expanding into diverse Python applications beyond the web. Currently learning Japanese (JLPT N5) and committed to growing as a developer through impactful, practical work.

Education

Dr. D Y Patil Institute of Engineering, Management & Research, Akurdi B.E. – Artificial Intelligence & Data Science	2023 – 2027 CGPA: 8.38 / 10
Z B Patil College, Dhule 12th – MSBSHSE	2023 Percentage: 81.17 / 100
Bohra Central School, Parola 10th – CBSE	2021 Percentage: 81.66 / 100

Internships

TechJar Technologies, Pune – Full Stack Web Developer 01 Sep 2025 – 30 Nov 2025
Key Skills: Python, Snowflake, Speech-to-Text (STT) Integration, CSS, GPT-based Conversational Systems, PowerBI, ReactJS, HTML, AI-Driven Workflow Automation, API Integration

- Completed a 3-month internship, contributing to InterviewBot and VDR (Virtual Data Room) projects.
- Developed React-based frontend features and supported backend development using Flask and Python libraries to build scalable APIs and application services.
- Integrated AI solutions including STT/TTS technologies (ElevenLabs, Deepgram, and other leading models), and conducted R&D to evaluate and recommend the most suitable speech models based on business requirements.
- Participated in on-ground testing and validation of InterviewBot; implemented secure file handling, role-based access control (RBAC), audit logging, data visualization dashboards, and OCR-based automation workflows for structured data extraction.
- Created a Streamlit application utilizing OCR and the Claude API to extract structured information from images of society register pages containing handwritten vehicle entries.

Projects

VIRA (Vigilance And Rapid Alerts) – Team Size: 6 25 Jul 2025 – 31 Mar 2026
Key Skills: Machine Learning, Reinforcement Learning, Web Development, .NET, FastAPI, Gymnasium, Isaac Sim, YOLO, FaceNet, WebSockets, PyTorch, TensorFlow, PyBullet
github.com/RohanSali/Drone-Surveillance-System

- Built an enterprise-grade surveillance platform integrating autonomous drones and fixed CCTV networks for real-time monitoring and AI-driven threat detection (facial recognition, crowd analysis, anomaly detection) across large geographic areas.
- Designed a real-time alert and response pipeline that geolocates incidents, computes optimal safe hover zones, and dispatches drones to validate alerts and enable rapid response in the absence of on-ground authorities.
- Engineered scalable, cloud-native infrastructure (deployed on Render) with real-time GPS tracking of multiple drones and zone-based surveillance management across regions.

Nihongono-Practice

14 Jan 2026 – 14 Feb 2026

Key Skills: React.js, CSS, Google Generative API, Flask, Google Firebase
github.com/aadim112/Nihongono-Practice

- Built a full-stack web application (ReactJS, CSS, Flask) to help learners practice Japanese reading, listening, vocabulary, grammar, and kanji, structured around JLPT proficiency levels.
- Integrated the Google Generative API to power interactive features, including conversational listening exercises with spoken-response practice and reading comprehension with auto-generated passages and questions.
- Developed dedicated modules for vocabulary management (custom dictionary), grammar pattern study by JLPT level, and kanji recognition practice, delivering a comprehensive, exam-oriented learning environment.

Pure Change – Team Size: 2

09 Oct 2025 – 07 Dec 2025

Key Skills: React.js, CSS, HTML5, Express.js, Firebase, Google Generative AI, Firestore, Firebase RTDB
github.com/aadim112/PureChange

- Built a production-grade web platform combining behavioral logic, AI personalization, and gamification to help users break destructive habits and build consistent daily routines.
- Designed a habit-tracking system with streaks, behavioral scoring, and league-based progression, alongside personalized AI-driven routines for yoga, meditation, fitness, and productivity with real-time feedback for urge control.
- Implemented engagement features (leaderboards, community chatrooms, referrals, contextual motivational content), secure authentication with role-based access, and a tiered (Free/Pro/Elite) payment-gated subscription model.

Medilog – Team Size: 4

01 Mar 2025 – 08 Mar 2025

Key Skills: React.js, HTML5, CSS3, Flask, Google Authentication, OAuth 2.0, Video Consultation APIs, Git, GitHub, OCR, Google Generative API
github.com/aadim112/HM0055_Code200

- Developed a full-stack hospital management application (React.js, Flask) with Google OAuth-based authentication and role-specific access for doctors, patients, and staff, supporting appointment booking, record management, and real-time video consultations.
- Built OCR-based digitization of prescriptions and medical reports into structured digital records, and integrated the Google Gemini API to analyze patient data and support clinical insights.
- Enabled secure patient access to consultation history and reports, doctor-side progress tracking across visits, and inter-hospital collaboration for resource sharing, plus AI-based inventory and equipment demand forecasting.

Publications / Research / White Papers

Transformers And Their Working Simplified

Kaggle

Key Skills: Large Language Models (LLMs), Self-Attention, Encoder-Decoder Architecture

- Developed an educational deep learning notebook explaining the Transformer architecture from first principles, covering self-attention, multi-head attention, positional encoding, encoder-decoder architecture, tokenization, embeddings, and LLM foundations, supported by visualizations and practical examples.
- Link: kaggle.com/code/aaditya112/transformers-and-their-working-simplified

Achievements

- Google Cloud Arcade – Novice

Assessments / Certifications

- **Python (Basics)** – HackerRank (*Python, OOP, Data Structures, Debugging*)
- **Core Concepts on C++** – DevTown (*C++, OOP, Data Structures, Encapsulation, Abstraction, Polymorphism, Classes & Objects, Inheritance*)
- **Introduction to Programming Using HTML and CSS** – SoloLearn (*HTML5, Web Development*)
- **Core Concepts of JavaScript** – SoloLearn (*JavaScript, DOM Manipulation, Event Handling, Functions, Objects, Conditional Statements, Web Development, Interactive User Interfaces*)
- **Deloitte Australia – Data Analytics Job Simulation** – Forage (*Data Analytics, Business Analytics, Data Interpretation, Data-Driven Decision Making, Microsoft Excel*) – Completed a virtual job simulation focused on data analytics and forensic technology, gaining practical experience in analyzing datasets, identifying patterns, deriving business insights, and applying analytical techniques to real-world business scenarios.
- **Data Pre-processing for Machine Learning in Python** – Unstop (*Data Preprocessing, Data Cleaning, Feature*

Engineering, NumPy, Pandas, One-Hot Encoding, Data Integration, Data Reduction, Outlier Detection, Data Quality Management, ML Pipeline)

- **Machine Learning Crash Course** – Google Developer Program (*Machine Learning, Supervised Learning, TensorFlow, Data Preprocessing, Feature Engineering, Classification, Regression, Neural Networks, Deep Learning Fundamentals*) – g.dev/aditya149

Positions of Responsibility

Joint Competitive Programming Head – CSI

- Served as the Competitive Programming Head at CSI, responsible for planning and managing all activities related to competitive programming.
- Successfully organized and conducted a coding competition with participation from around 10 teams, where participants solved medium-level programming challenges using Python or C++.
- Contributed to the club's broader initiatives by supporting and coordinating various events and activities.

Co-Curricular Activities

- Led a team of 4 for the 24-hour InnoHack Hackathon organized by PCCOE College; ranked in the top 10 among 300 teams.
- Participated in SIH 2025 Hackathon, presenting and demonstrating the VIRA prototype.
- Selected in the top 10 teams among 500 for the CodeAlpha Hackathon organized by VIT.
- Led a team of 4 for a PBL presenting a working model of a Smart Home System developed using Arduino.

Extra-Curricular Activities

- Working as a Graphics Designer on Fiverr since 2022; served 30+ international clients and delivered 30+ orders.
- Completed the 6-month Google Arcade program, gaining hands-on experience with Google Cloud Platform and reaching the Arcade Novice tier.

Personal Details

Gender:	Male
Marital Status:	Single
Date of Birth:	30 Jun 2005
Known Languages:	English, Marathi, Hindi, Japanese
Current Address:	Navashakti Society, Ravet, Pimpri-Chinchwad, Pune, Maharashtra, India – 412101
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Web Links

- Personal / Portfolio: aadim112.github.io/aaditya
- GitHub: github.com/aadim112
- Codolio: codolio.com/profile/Aaditya